THE UNIVERSITY OF BRITISH COLUMBIA
Psychology
Irving K. Barber Faculty of Arts and Social Sciences
Okanagan Campus

Participant Information and Consent Form

Title of Study: Testing motor system engagement across the action continuum

Principal Investigator(s): Sarah Kraeutner, PhD, Assistant Professor, University of British Columbia, Okanagan (sarah.kraeutner@ubc.ca; 250 807 8823)

Study Team:
Timothy Welsh, University of Toronto, Toronto;
Celine Balay, University of British Columbia, Okanagan
AJ Neil, University of British Columbia, Okanagan;
Sarvenaz Heirani Moghaddam, University of British Columbia Okanagan;
Justine Magnuson, University of British Columbia Okanagan

Study Site: The Neuroplasticity, Imagery, and Motor Behaviour Laboratory, ART 280L, University of British Columbia – Okanagan

Invitation

You have been invited to take part in this research study. A research study is a way of gathering information on a treatment, procedure, or medical device or answering a question about something that is not well understood. You are being asked to take part in this study because you replied to our advertisement, you meet the study requirements, and you are free of any brain injury or disease and meet the inclusion criteria for the study.

Your participation is voluntary

Your participation is voluntary. You have the right to refuse to participate in this study. If you decide to participate, you may still choose to withdraw from the study at any time without any negative consequences to the medical care, education, or other services to which you are entitled or are presently receiving.

Please review the consent document carefully when deciding whether you wish to be part of the research and sign the consent form only if you accept being a research participant.

The researchers will:

- Discuss the study with you

- Answer your questions
- Keep confidential all information which could identify you personally
- Be available during the study to deal with problems and answer questions

Who is conducting this study?

This study is being conducted by the Neuroplasticity, Imagery, and Motor Behaviour Laboratory, funded by the Natural Sciences and Engineering Research Council of Canada (NSERC).

Background

When we intentionally move, our brain sends signals to our muscles. When we imagine or observe movement (called 'action states'), we experience a phenomenon called motor overflow. This means that we have a ‘spillover’ of the same brain signals that result in unintentional or micro movements in muscles. These micro movements are not noticeable to the individual imagining or watching. However, these micromovements reflect broader activation of the 'motor system': the parts of our brain and body that are required for us to move and perform actions.

While we know these micro movements exist when imagining or watching movements, we do not know a lot about *where* these micromovements arise from between the brain and muscles, or whether the two types of action states involve the motor system differently. We are recruiting 80 participants for this study.

What is the purpose of the study?

This study will examine whether the signals from the brain to the muscles differ across action states, like imagination and observation of a movement. To do this, we will use tools that allow us to measure brain activity and muscle activity, to detect both the amount of movement-related brain activity and the strength of the signals reaching the muscles. This will help us better understand how the motor system contributes to different action states.

Who can participate in this study?

Inclusion Criteria

1. Adult 19 or over
2. Normal or corrected to normal vision

3. Fluent in spoken and written English

Who cannot participate in this study?

Exclusion criteria:
1) Injuries, 2) movement disorders, or 3) neurological conditions that make coordinated movements of your hand and arm difficult or painful

I. STUDY PROCEDURES

If you agree to participate in this study, some demographic information such as your age and gender, will be collected on study data forms to help us characterise our study sample, and you will be asked to complete a questionnaire that measures how easily you are able to imagine movements. This information will help us understand how easily or difficult you find practicing using motor imagery. This questionnaire should take you approximately 10-15 minutes.

Behavioural Task

You will engage in a task where you are asked to perform side-to-side tapping movements with your arm and hand on a touchscreen computer (i.e., moving from a 'left' target to a 'right' target). You will perform this task in four different ways: execution (physical movement), imagination of the movement focusing on the sensations associated with the movement, imagination of the movement focusing on the picture of the movement, and observation of someone else performing the movement (via a video). During this task, you will be seated in front of a task with a touchscreen computer placed flat on the desk in front of you. You will be asked to rest your hands quietly on the desk. When performing the action, you will perform the action continuously for up to 20 seconds. Sometimes you will also be asked to rest quietly while staying still. At the outset of the task, you will be familiarized with the movement, and oriented to the different action states (execution (physical movement), imagination of the movement focusing on the sensations associated with the movement, imagination of the movement focusing on the picture of the movement, and observation of someone else performing the movement (via a video).

Periodically throughout the task (after each block), you will also be asked to complete two short questionnaires that measure attention and task engagement, and mental load or effort. Overall,

the behavioural task will take up to 35 minutes to complete and you will be provided with breaks throughout.

Muscle and Brain Activity

During the task, we will measure engagement of the motor system. To measure your muscle activity, we will use a technique called surface electromyography (EMG). Prior to beginning the task, we will place electrodes (like stickers) on muscles on both of your hands and both of your arms. Because the electrode 'stickers' are placed on your arm and hand, it is best if you wear a short-sleeved shirt. If you are wearing a long sleeved shirt, you may be asked to roll up your sleeves.

You may also be asked to wear an electrode cap (like a net) with sensors affixed to it. This cap allows us to measure brain activity, signals produced by your brain, using an electroencephalogram (EEG). Measuring brain activity using EEG is passive and painless. It does not send any signals into your body - recording this activity allows us to 'listen' to the natural signals your brain produces. The electrode cap is wet, soaked in salt water, so your hair will get wet.

Experimental procedure

Upon the single visit to the lab lasting 50-80 minutes, you will first complete the initial questionnaires. You will then be set up to complete the task. You will then be asked to perform the task in the different action states and provided with rest throughout.

Voluntary Participation

Participating in this research study is completely voluntary. Even if you decide to participate now, you may change your mind and stop at any time for any reason, without penalty or loss of compensation to which you are entitled. If you decide not to take part in the study or if you leave the session early, your data will be automatically withdrawn from the study. Once you complete the session, your data can no longer be withdrawn from the study, as this data is automatically added to the database and entered into the analysis.

II. **STUDY RESULTS**

If you would like a description of the results at the end of the study, you can obtain a short description of these results by visiting theNIMBL.com in approximately 12 months. It is anticipated that the results of this study will be published and/or presented in a variety of forums, including

academic journal articles and presentations. In any publication and/or presentation, information will be provided in such a way that you cannot be identified. Specifically, no individual results will be provided, and only de-identified data will be associated with the study results. If a publication results, a preliminary copy of this publication will also be accessible by contacting the principal investigator of this study: Dr. Sarah Kraeutner (sarah.kraeutner@ubc.ca; Tel: 250-807-8823)

III. POTENTIAL RISKS OF THE STUDY

What are the possible harms and discomforts?

Potential Risks Associated with surface EMG

Very Rare (<0.01%): There is minimal risk related to the use of this technique. The electrodes lie on top of the skin (like a sticker on your skin) and a conductive gel provides the contact between the skin and the electrodes. In uncommon instances (1 or more out of every 10,000 people but less than 1 out of every 1000 people) it is possible that your skin may be sensitive to the conductive gel, alcohol or adhesive used in the application of the electrodes. In such cases a rash or reddening of the skin is possible. This usually goes away in 2-3 hours after removing the sensors.

Potential Risks Associated with Behavioural Tasks

Very common (53%): The risks associated with behavioural tasks (like the music listening task) are minimal; you may become bored or fatigued from participating in this research.

However, you will be given breaks throughout the behavioural task to reduce these risks.

Potential Risks Associated with EEG:

EEG involves a 'cap' (like a net) that has been soaked in salt water placed on the head. The salt water or the cap itself (particularly where the electrodes touch the scalp) may cause mild, temporary skin redness or itching or irritation. EEG also requires you to stay still. Sitting still for 30 to 60 minutes can make participants' neck or back feel stiff. To mitigate stiffness, breaks have been embedded throughout the task and overall experimental session where you can stretch or move.

There may be other risks that have not yet been identified, and unexpected side effects that have not been previously observed may occur.

IV. POTENTIAL BENEFITS OF THE STUDY

There may be no direct benefits to you from taking part in this study.

This study has the potential to benefit society through the generation of knowledge related to our understanding of different action states. We hope that the information learned from this study may be useful to enhance learning through repeated imagined practice of a task, or observation of a task, in applications including sport, surgical skills, or neurorehabilitation.

V. CONFIDENTIALITY

How will your privacy be protected?

Your confidentiality will be respected. However, research records may be inspected in the presence of the Investigator or designate and by the representatives of UBC's Clinical Research Ethics Board for the purpose of monitoring the research. No information or records that disclose your identity will be published, nor will any information or records that disclose your identity be removed or released without your consent unless required by law.

You will be assigned a unique study number as a participant in this study. This number will not include any personal information that could identify you (e.g., it will not include your Personal Health Number, SIN, initials, etc.). Only this number will be used on any research-related information collected about you during the course of this study, so that your identity will be kept confidential. Information that contains your identity will remain only with the Principal Investigator and/or designate. The list that matches your name to the unique study number that is used on your research-related information will not be removed or released without your consent unless required by law.

Your rights to privacy are legally protected by federal and provincial laws that require safeguards to ensure that your privacy is respected. You also have the legal right of access to the information about you, and, if need be, an opportunity to correct any errors in this information. Further details about these laws are available on request to the study team.

Data records will be kept on the NIMBL computer hard drive. The security of the hard drive is password protected. Only Dr. Kraeutner and members of her research team are able to access these drives. Your study-related information and data will be used at UBCO only by Dr. Kraeutner,

and members of the research team who are listed on this consent form. If the results of this study are published or presented in public, information that identifies you will be removed. If you decide not to sign the form, you cannot participate in the study.

Open Access At Publication

Researchers may be required by granting agencies and journals to make their findings and/or de-identified data available at the time of publication. Your de-identified research data may be published or deposited into a publicly accessible location at the time of publication. This means that data will be stored in a publicly accessible repository and other researchers may analyze the data for different reasons other than those described in this consent form. At no time will identifying information, such as your name, birth date or street address be included in such data. Once the data is made publicly available, you will not be able to withdraw your data. We acknowledge that open access to de-identified data has the potential to increase participant risk, as others may be able to link the data to you or use it for unintended purposes."

Disclosure of Race/Ethnicity

Studies involving humans now routinely collect information on race and ethnic origin as well as other characteristics of individuals including gender identity to help characterise the sample and ensure inclusive and accurate representation of the study population. You should be aware that providing this information is not mandatory.

What happens if something goes wrong?

By signing this form, you do not give up any of your legal rights and you do not release the principal investigator, participating institutions, or anyone else from their legal and professional duties. If you become ill or physically injured as a result of participation in this study, medical treatment will be provided at no additional cost to you. The costs of your medical treatment will be paid by your provincial medical plan.

What will the study cost me?

All research-related procedures that you will receive during your participation in this study will be provided at no cost to you.

Compensation/remuneration

You will receive $10 OR 1.5 total SONA credits if you participate through the SONA system for giving your time and/or to offset your parking and/or travel expenses incurred to participate in this study. This compensation is intended as an honorarium — a gesture of appreciation for volunteering your time — and not as a form of employment or fee for service.

VI. CONTACT FOR INFORMATION ABOUT THE STUDY

Who can you contact if you have questions about the study?

If you have any questions or desire further information about this study before or during participation you can contact Dr. Sarah Kraeutner at 250-807-8823 or sarah.kraeutner@ubc.ca.

Who can you contact if you have concerns or complaints about the study?

If you have any concerns or complaints about your rights as a research participant and/or your experiences while participating in this study, contact the Research Participant Complaint Line in the UBC Office of Research Services at 1-877-822-8598 or the UBC Okanagan Research Services Office as 250-807-8832. It is also possible to contact the Research Participant Complaint Line by email (RSIL@ors.ubc.ca). Please reference the study number [H26-01811] when contacting the Complaint Line so the staff can better assist you

VII. SIGNATURES

Study Title: Testing motor system engagement across the action continuum

Principal Investigator: Dr. Sarah Kraeutner

Participant Consent

My signature on this consent form means:

- I have read and understood the information in this consent form.
- I have been able to ask questions and have had satisfactory responses to my questions.
- I understand that my participation in this study is voluntary.
- I understand that I am completely free at any time to refuse to participate or to withdraw from this study at any time.
- I understand that I am not waiving any of my legal rights as a result of signing this consent form.
- I understand that there is no guarantee that this study will provide any benefits to me.

Participant's Signature	Printed Name	Date

Signature of Person Obtaining Consent	Printed Name	Study Role	Date